

sparsity_jac

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This file is part of CasADi.

CasADi -- A symbolic framework for dynamic optimization.
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1 sparsity_jac

```
[1]: import casadi as ca
      from numpy import *
      import casadi as c
      from pylab import spy, show
```

We construct a simple SX expression

```
[2]: x = ca.SX.sym("x",40)
      y = x[:-2]-2*x[1:-1]+x[2:]
```

Let's see what the first 5 entries of y look like

```
[3]: print(y[:5])
```

```
(((x_0-(2.*x_1))+x_2), ((x_1-(2.*x_2))+x_3), ((x_2-(2.*x_3))+x_4),
((x_3-(2.*x_4))+x_5), ((x_4-(2.*x_5))+x_6)]
```

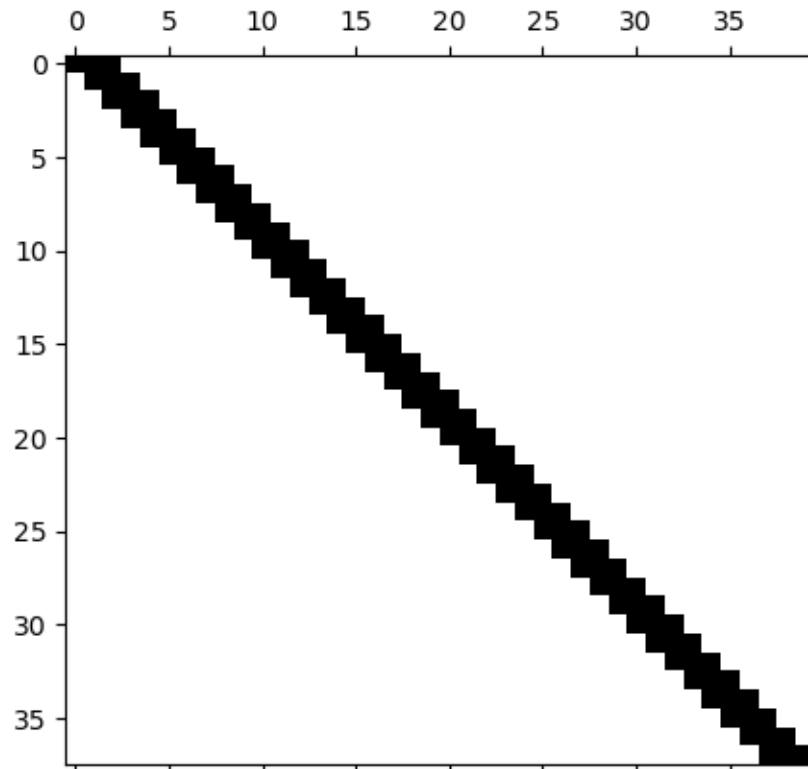
Next, we construct a function

```
[4]: f = ca.Function("f", [x],[y])
```

And we visualize the sparsity of the jacobian

```
[5]: spy(f.sparsity_jac(0, 0))
```

```
[5]: <matplotlib.image.AxesImage at 0x7f56191b39a0>
```



```
[6]: show()
```